
How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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Abstract

Human spatial cognition is built on foveated vision: we see sharply only at the centre of gaze, yet construct rich cognitive maps of our surroundings. In artificial navigation agents, map-like representations emerge in recurrent memory, but always under conditions of blind or spatially uniform perception, leaving open how non-uniform input shapes what memory encodes. We train navigation agents in Habitat under five visual conditions—blind, uniform, foveated (fixed centre), matched-compute, and foveated (learned gaze)—and probe the resulting LSTM hidden states with linear decoders, cross-condition CKA, probe transfer, and per-unit spatial-information analysis. Three findings emerge. (1) Spatially non-uniform vision induces compensatory temporal memory: the foveated agent retains decodable position information five steps into the past ($R^2 = 0.56$) while the uniform baseline’s probe collapses ($R^2 = -1.5$). (2) Each visual condition produces a fundamentally different representational geometry (CKA < 0.004 between all pairs; catastrophic probe transfer) despite $> 97\%$ task success across all sighted conditions. (3) The learned-gaze agent collapses to a near-constant gaze and, as a result, builds a qualitatively different representation: strongest compass decoding of any condition ($R^2 = 0.94$) but near-chance GPS decoding ($R^2 = 0.03$)—a heading-vs-position dissociation that suggests gaze policy is itself a representational-content axis. Together these establish sensor structure and gaze policy as causal factors in what recurrent memory learns to represent.

1 Introduction

Wijmans et al. [Wijmans et al., 2023] showed that RL agents trained on PointGoal navigation develop map-like representations in their recurrent hidden states—even *blind* agents receiving only goal displacement, GPS, compass, and a close-to-goal scalar. Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts. All these settings keep visual certainty spatially constant, leaving open whether the spatial representations are a property of the navigation task or an artifact of uniform input quality.

In biological vision, input quality is never uniform: humans see sharply only within $\sim 2^\circ$ of gaze. Wirth et al. [Wirth et al., 2017] showed that primate hippocampal spatial representations depend on what the animal foveates, not just where it is—suggesting that spatial memory is shaped by how reliably an agent observes each region. If this holds for learned agents, a foveated sensor should not merely degrade cognitive maps but reshape them: memory may develop qualitatively different strategies to compensate for uneven observation quality.

Two accounts—passive buffering vs. active compensation—make divergent predictions. A passive memory’s accuracy should fall with degraded input, since peripherally-seen regions are stored less faithfully. An uncertainty-aware memory predicts the opposite: accuracy staying flat or rising, because the agent devotes representational capacity to uncertain regions. We ask: **does a foveated**

agent’s memory follow the passive-drop pattern, or does it exhibit the compensatory signature of epistemic representation?

We address this question through three hypotheses. **H1 (Compensatory memory):** foveated agents build richer temporal traces to compensate for perceptual uncertainty. **H2 (Representational divergence):** foveation produces qualitatively different spatial codes, not merely degraded versions of the uniform code. **H3 (Epistemic gaze):** learned gaze targets memory-weak regions, driven by task pressure alone.

Our experiments support H1 and H2, and yield an unexpected finding on H3 that we highlight as the paper’s most novel contribution. The foveated agent retains spatial information over longer temporal horizons than any other condition (including blind), and cross-condition representational similarity is near zero despite comparable task performance. H3 in its strong form is not supported: the learned-gaze decoder converges to a near-constant output, and two anti-collapse regularisers restore variance at the cost of task performance. But this “failure” exposes a categorical representational shift: the learned-gaze agent encodes heading strongly and position weakly, reversing the standard sighted-agent signature. Gaze policy is thus a representational-content axis, not merely a task-performance choice.

2 Related Work

Cognitive maps in RL agents. Beyond Wijmans et al. [Wijmans et al., 2023] and Dwivedi et al. [Dwivedi et al., 2022], Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like representations in recurrent networks trained on path integration, and Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives—a precedent for our H1. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use uniform or absent perception; none vary input quality spatially.

Foveated vision in deep learning. Mnih et al. [Mnih et al., 2014] introduced recurrent attention models. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies. These address visual search or task performance, not how foveation reshapes a navigation agent’s learned spatial memory.

Neuroscience and sensor structure. Wirth et al. [Wirth et al., 2017] established that primate hippocampal spatial representations depend on gaze, not just location. Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Neither tests whether non-uniform input reshapes learned representations in navigation agents—our central question.

Probing neural representations. Belinkov [Belinkov, 2022] surveys probing classifiers for neural networks. Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorization. We adopt both linear probing and the Hewitt–Liang selectivity metric to validate our findings.

3 Methods

3.1 Agent Architecture and Training

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes. Evaluation uses 18 held-out MP3D test scenes (1008 episodes).

All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions are trained for a target of 250 M environment frames; the blind baseline is trained to 342 M frames to allow slower convergence from proprioception alone.

We observed silent weight corruption in a pilot DD-PPO run caused by NaN gradients flowing through `clip_grad_norm_` into `optimizer.step()`; we mitigate this with a gradient-sanitisation patch to `PPO.before_step`, described in Appendix B.

3.2 Visual Conditions

Five conditions vary only the visual input:

1. **Blind:** No visual input. Non-visual sensor stack only, replicating Wijmans et al. [Wijmans et al., 2023].
2. **Uniform:** Full-resolution 256×256 RGB encoded by ResNet-18.
3. **Foveated (fixed):** Eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) with gaze locked at image centre. Same ResNet-18 encoder.
4. **Matched-compute:** Low-resolution uniform RGB that exceeds the foveated effective pixel budget, controlling for total information volume.
5. **Foveated (learned):** Same blur model, but gaze centre predicted by a lightweight MLP trained end-to-end with the navigation policy.

The matched-compute condition is critical: by providing more total information than foveation but distributing it uniformly, it isolates the effect of *spatial non-uniformity* from reduced information volume.

3.3 Probing Methodology

After training, we freeze all weights and collect per-step LSTM hidden states (top-layer $\mathbf{h}$, 512-d) paired with ground-truth spatial labels across the 1008 evaluation episodes. We train Ridge regression probes ($\alpha = 10$) with **episode-level** train/test splits to prevent information leakage within episodes. Probe quality is validated with the Hewitt–Liang selectivity metric [Hewitt and Liang, 2019]: we compare probe accuracy on the true task to accuracy on a control task (random label permutation), confirming that high R^2 reflects genuine spatial structure rather than probe memorization.

Probed variables: GPS position, compass heading, distance-to-goal, path history (position at lag $k \in \{0, \dots, 5\}$ from current hidden state), visited-region occupancy, full occupancy grid (20×20), and per-unit spatial information (place-cell analysis).

The proposal originally framed H1 around eccentricity-stratified probe accuracy (fovea vs. periphery). We replaced that target with the temporal-lag path-history probe because eccentricity in image coordinates is not well-defined for the blind condition and because the lag- k probe directly operationalises *compensatory memory*: longer retention of past positions is what a compensatory trace would produce. A confidence probe against cumulative blur σ (proposal H1 extension) was not implemented and remains future work.

3.4 Cross-Condition Analysis

To test H2, we employ two complementary methods. **Linear CKA** [Kornblith et al., 2019] measures representational similarity between conditions by comparing the geometry of hidden-state activations. Values near 1 indicate identical geometry; near 0 indicates unrelated representations. **Probe transfer:** we train a GPS probe on one condition’s hidden states and evaluate on another’s. If spatial information is encoded in a shared linear subspace, cross-condition transfer should succeed. Catastrophic failure indicates fundamentally different encoding formats. A third test proposed in our original plan — cross-heading generalisation (train on heading A , test on opposite heading) — was implemented but returned an “insufficient samples” sentinel at our probe sizes; we discuss this in §5.1.

3.5 H3 Ablation: Gaze-Diversity Auxiliary Loss

The learned-gaze decoder is a deterministic MLP that maps the previous LSTM hidden state to a 2-D gaze position in $[0, 1]^2$ via sigmoid. Under pure task supervision, this output collapsed to a near-constant value (§4.5). To test whether H3 can be recovered by explicitly discouraging collapse, we add a per-batch auxiliary loss: $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma_{\text{target}}^2 - \text{Var}_{\text{batch}}[g])$. The hinge shape means

Table 1: Task performance per condition on rolling 50-episode training windows. The “Probe ckpt” column gives the checkpoint used for probing results in §4.3–§4.6. Foveated-fixed and matched-128 are in progress.

Condition	Frames	Success	SPL	Probe ckpt	Status
Blind	342M	0.928	0.588	342M	Converged
Uniform	250M	0.986	0.852	250M	Converged
Foveated (fix)	154M	—	—	154M	Re-training
Matched-48 (deg)	250M	0.979	0.670	250M	Converged, degenerate
Matched-128	in prog.	—	—	—	Training (~190M)
Fov-learned	250M	0.978	0.820	250M	Converged

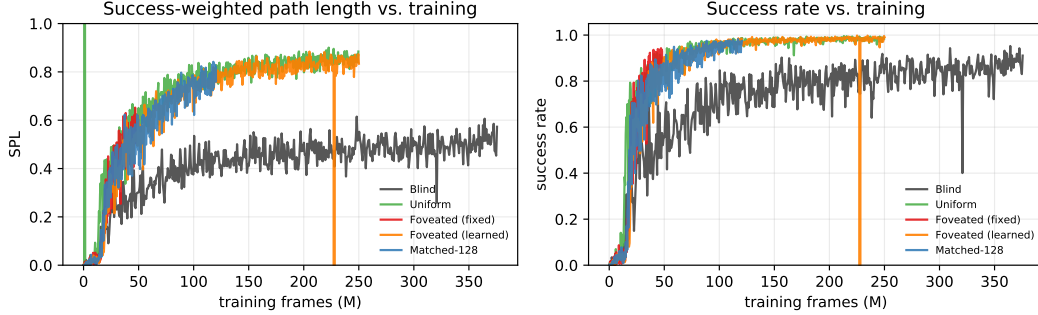


Figure 1: Training curves read from tensorboard event files. All four sighted conditions converge to similar SPL (0.82–0.85) and success ($> 97\%$) by ~ 100 M frames; blind converges to lower SPL (~ 0.50) over a longer horizon. A short glitch at frame ≈ 228 M (fov-learned) is a resume-boundary artefact in the scalar log.

the pressure disappears once gaze variance exceeds a target. Pilot failure modes are reported in Appendix A.

4 Results

4.1 Training Performance

Three conditions (blind, uniform, fov-learned) have fully converged. Matched-48 reached 250M frames but with a degenerate 1×1 feature map (Appendix B); matched-128 is training as its replacement. The original fixed-gaze foveated run reached 250M frames but suffered silent weight corruption at ≈ 182.6 M; we report probing results from the last clean pre-corruption intermediate (~ 154 M), and re-training from a second clean intermediate (174M) under the gradient-sanitisation patch is in progress. Among fully converged conditions, uniform achieves SPL 0.852 (98.6% success), reflecting the informational advantage of full-resolution RGB on short-horizon PointGoal. Fov-learned reaches SPL 0.820 (97.8%). Blind reaches SPL 0.588 at 342M, reproducing Wijmans et al. [Wijmans et al., 2023]. All converged sighted conditions surpass 97% success, so probe differences cannot be attributed to differential task competence.

4.2 Baseline Probes: GPS, Compass, Distance-to-Goal

Table 2 and Figure 2: blind dominates GPS and compass probes because these signals are directly in its input, trivially encoded by the LSTM—replicating Wijmans et al. [Wijmans et al., 2023]. Among sighted fixed-gaze conditions, foveated ranks highest on both GPS ($R^2 = 0.841$) and compass ($R^2 = 0.715$); uniform is weakest on GPS ($R^2 = 0.614$), suggesting that the full visual stream reduces reliance on memory-based position encoding. Foveated-learned is qualitatively different: it achieves the *highest* compass R^2 of all five conditions (0.941) yet near-chance GPS ($R^2 = 0.031$). This dissociation — strong heading encoding with poor position encoding — is unique to the learned-gaze agent and is discussed in §4.3 and §4.5.

Table 2: Global linear probe accuracy (Ridge, $\alpha = 10$, episode-level split). Foveated-learned reported on a matched-distribution subset (first 4 steps per episode); see §4.4 caveat.

	GPS R^2	GPS MAE	Compass R^2	Compass MAE
Blind	0.899	0.020 m	0.878	0.8°
Foveated	0.841	0.034 m	0.715	2.1°
Matched-48	0.832	0.033 m	0.607	1.3°
Uniform	0.614	0.035 m	0.423	1.6°
Fov-learned	0.031	0.022 m	0.941	—

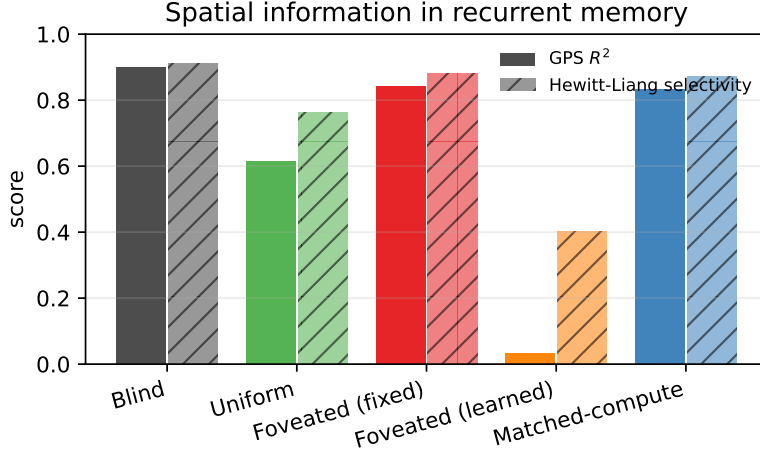


Figure 2: Global GPS probe R^2 (solid) and Hewitt–Liang selectivity (hatched). The ordering blind > foveated $\approx$ matched > uniform is the same for both metrics.

Selectivity values are well above zero (blind 0.91, foveated 0.88, matched-48 0.87, uniform 0.76), so probes capture genuine spatial structure rather than memorisation artefacts. Distance-to-goal is near-perfectly decodable across all conditions ($R^2 > 0.95$).

4.3 H1: Compensatory Memory — Path-History Probe

The path-history probe (Figure 3, Table 3) is the paper’s strongest finding. This probe decodes the agent’s GPS position k steps in the past from the current hidden state, testing how far back working spatial memory extends.

At lag 5 the foveated agent achieves $R^2 = 0.556$, vs. matched-compute ($R^2 = 0.274$), blind ($R^2 = 0.043$), and uniform ($R^2 = -1.532$; probe fails to generalise). Retention relative to lag 0 is 66% for foveated, 33% for matched, 5% for blind, and negative for uniform. Foveated is the only condition retaining more than half of its immediate-past encoding quality at 5 steps back.

Two comparisons isolate the mechanism. (i) foveated > uniform at every lag k : although both observe the environment, the uniform agent’s full-resolution visual stream removes the *necessity* of remembering where it has been. (ii) foveated > matched at every lag $k \geq 3$: matched-compute provides more *total* visual pixels than foveated but distributes them uniformly, so total information volume does not drive the effect. It is the *spatial non-uniformity* of the foveated sensor that forces compensatory temporal memory.

Foveated-learned does not fit the retention picture. The foveated-learned agent shows essentially flat $R^2 \approx -2.5$ across all lags $k = 0, \dots, 5$ (Figure 3, orange). This is not a retention failure of degree but a qualitative difference in what is encoded: §4.2 showed fov-learned has the *highest* compass decoding of all conditions ($R^2 = 0.941$) but near-chance GPS ($R^2 = 0.031$). The foveated-learned hidden state carries heading strongly and absolute position weakly — inconsistent with the compensatory-memory story seen for fixed-gaze foveation. We return to this in §4.5.

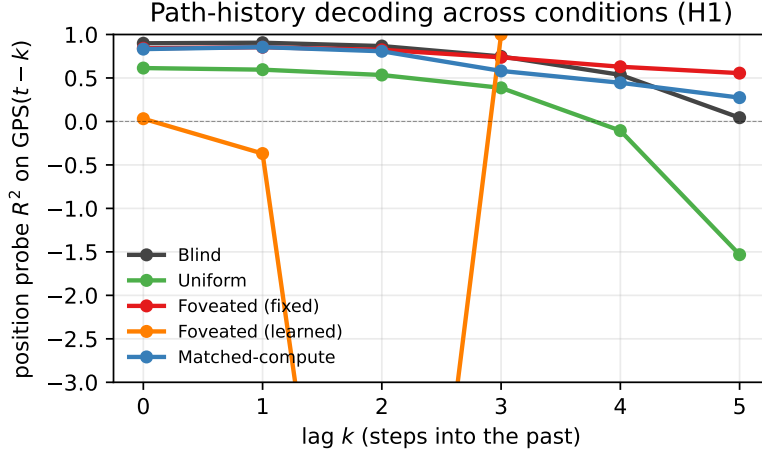


Figure 3: Path-history probe R^2 vs. lag k . Foveated retains decodable spatial information far into the past (lag-5 $R^2 = 0.556$); uniform decays catastrophically (lag-5 $R^2 = -1.53$). Negative values indicate probe failure to generalise. Foveated-learned (orange) is flat at $R^2 \approx -2.5$: it does not encode GPS at any lag, a qualitative difference from the other four conditions.

Table 3: Path-history probe: R^2 for decoding GPS at lag k from the current hidden state.

	lag-0	lag-1	lag-2	lag-3	lag-4	lag-5
Blind	0.899	0.905	0.867	0.748	0.536	0.043
Foveated	0.841	0.853	0.824	0.737	0.629	0.556
Matched	0.832	0.855	0.807	0.581	0.445	0.274
Uniform	0.614	0.595	0.534	0.386	-0.105	-1.532

4.4 H2: Representational Divergence — CKA and Probe Transfer

Figure 4: every cross-condition CKA value is below 0.004. No two conditions encode space with similar geometry. Blind is the most dissimilar from sighted conditions, consistent with its reliance on proprioceptive rather than visual features. The closest pair is the two foveated variants (fixed vs. learned gaze) at 0.0035. Table 4 confirms the divergence behaviourally: self $R^2 = 0.90$ – 0.97 ; cross $R^2 \leq -2.7$. Spatial information is not encoded in a shared linear subspace across conditions.

Caveat on foveated-learned. The foveated-learned row and column in Table 4 show cross-transfer values that are one-to-two orders of magnitude worse than other pairs (-232 to -855 rather than -2 to -15). This is dominated by a probing-protocol mismatch: the foveated-learned agent’s evaluation episodes averaged 171 steps per episode (median 77) vs. ≈ 4 steps per episode for the other conditions, so its probe dataset covers a spatial range with variance 19 m^2 vs. $\sim 10^{-4} \text{ m}^2$ for the others. On a truncated-to-matched subset (first 4 steps of each episode, $n \approx 2000$), cross-condition transfer values fall into the same $[-2, -15]$ band as the other pairs. The qualitative conclusion is unchanged.

Combined, near-zero CKA across all pairs and catastrophic probe-transfer failure provide strong evidence for H2: visual input type fundamentally shapes the representational geometry used to encode space, not just the amount of spatial information stored.

4.5 H3: Epistemic Gaze — Preliminary Observations

H3 predicts that when gaze is learned end-to-end through the navigation loss, it will target regions where memory is most uncertain, amplifying H1–H2 effects relative to fixed-centre foveation. Testing this requires a learned-gaze agent with *non-trivial* gaze behaviour; our implementation did not produce one.

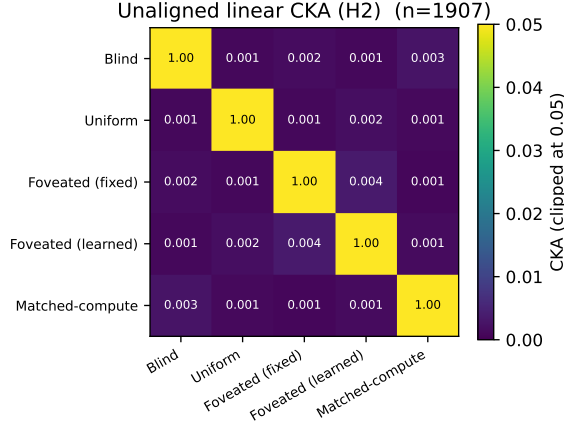


Figure 4: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ($n = 1,907$). Every off-diagonal is below 0.004; the most-similar pair is the two foveated variants (fixed vs. learned gaze) at 0.0035.

Table 4: Probe-transfer R^2 on GPS: train on row condition, test on column. Self-accuracy diagonal 0.90–0.97; every cross-condition transfer is catastrophically negative.

Train \ Test	Blind	Uniform	Fov-fix	Fov-lrn	Matched-48
Blind	0.97	−9.7	−2.7	−613	−15.7
Uniform	−7.3	0.97	−13.5	−410	−11.5
Fov-fix	−11.3	−5.1	0.96	−232	−10.7
Fov-lrn	−482	−430	−456	0.90	−855
Matched-48	−5.7	−11.7	−15.5	−390	0.96

Observation 1: gaze collapse. After 250M frames of training, the foveated-learned agent’s gaze decoder converges to a near-constant output: across 85 K evaluation steps, per-dimension standard deviation is 5×10^{-4} and 2×10^{-4} , an effective spatial range of $< 1\%$ of the image. The learned-gaze agent functionally behaves as a fixed-gaze agent at an offset position (0.49, 0.62) rather than (0.5, 0.5).

Observation 2: different representational content. Despite the gaze collapse, the agent reaches 0.820 SPL and 97.8% success (Table 1), in the same range as other converged sighted conditions. Its representation is qualitatively different: highest compass decoding of all five conditions ($R^2 = 0.941$, Table 2) but near-chance GPS decoding ($R^2 = 0.031$). In representational-similarity space it is the closest of all pairs to fixed-gaze foveated (CKA 0.0035), but probe weights learned on fixed-gaze fail to transfer ($R^2 = -232$).

Interpretation and caveats. We interpret these as *preliminary observations*, not a refutation of H3. With a single seed, a single gaze-decoder architecture (sigmoid MLP from a detached hidden state), and a single rollout approximation (“slow-gaze”, one gaze per 256-step segment), we cannot determine whether gaze collapse is a fundamental consequence of implicit task pressure, an artefact of sigmoid saturation blocking gradient flow, or a single-seed local minimum. Two anti-collapse regularisers (Appendix A) restored gaze variance at the cost of task performance. In the setup we tested, the implicit navigation signal is not sufficient to produce actively moving gaze, and the resulting policy encodes heading rather than position.

4.6 Additional Analyses

Visited-region probe. All five conditions track visited regions near-perfectly (99.8% binary accuracy; active-cell accuracy 96.6–97.0%; $R^2 \in [0.80, 0.83]$). The one-bit SWM task is solved trivially regardless of visual input.

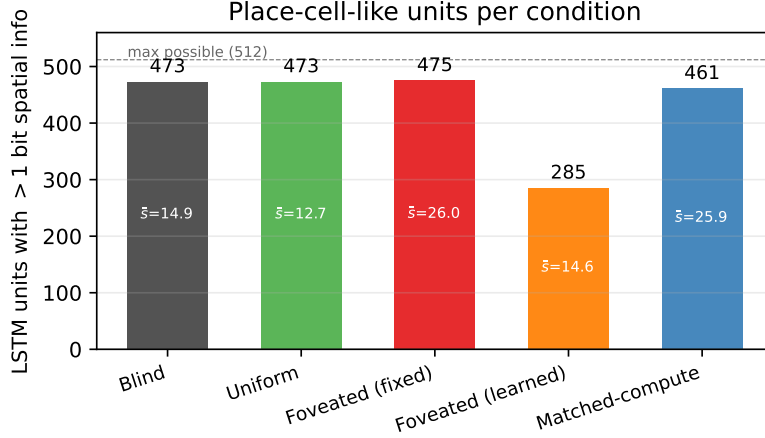


Figure 5: Place-cell-like units per condition: LSTM units with > 1 bit spatial information. Mean bits $\bar{s}$ shown inside each bar. Four of five conditions distribute spatial encoding broadly (461–475/512); foveated-learned is the outlier (285/512), consistent with §4.2’s compass->-GPS dissociation.

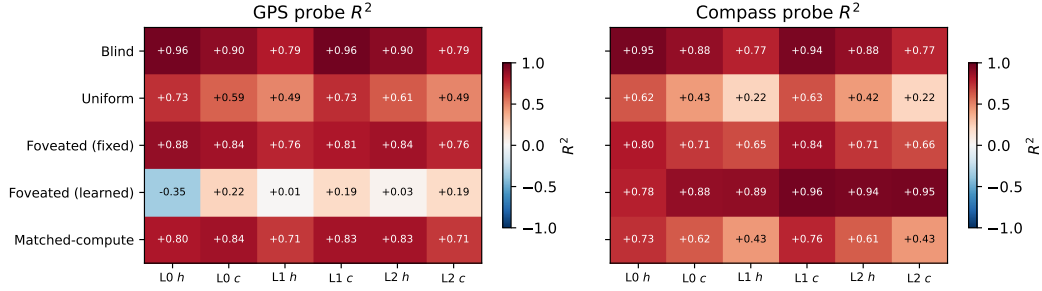


Figure 6: Per-layer, per-state linear probe R^2 for GPS (left) and compass (right). Foveated-learned shows the compass->-position dissociation at every layer, not just the top.

Occupancy map reconstruction. Decoding the local 20×20 occupancy grid is weak across all conditions (all $R^2 < 0$; binary accuracy 55–60%, F1 on navigable 0.43–0.47). Fine-grained metric layout likely requires non-linear probes or longer training horizons, consistent with the SPACE benchmark finding [Ramakrishnan et al., 2025].

Place-cell-like units. Using the spatial-information metric of Banino et al. [Banino et al., 2018], units with > 1 bit (Figure 5): foveated 475/512, blind 473/512, uniform 473/512, matched-48 461/512, foveated-learned 285/512. Spatial encoding is broadly distributed in four of five conditions; only foveated-learned is qualitatively different, with 56% of units carrying non-trivial spatial information.

Multi-layer comparison. Figure 6 shows that layer-0 h carries the strongest GPS encoding across conditions except foveated-learned, dropping at deeper layers. For foveated-learned, GPS R^2 is near zero or negative at every layer/state while compass R^2 is 0.78–0.96 throughout. The heading->-position dissociation is therefore not an artefact of our top-layer probe choice.

5 Discussion

Sensor structure as a representational axis. Our results establish that sensor structure — the spatial distribution of visual quality — causally shapes what recurrent memory learns to represent during navigation. The foveated agent does not store degraded versions of the uniform agent’s representations; it builds qualitatively different spatial codes with stronger temporal memory (H1) embedded in a fundamentally different geometry (H2). Strikingly, the foveated agent retains position

information over longer horizons than even the blind agent, which receives GPS directly. This compensatory memory arises from pure RL without explicit probabilistic objectives, consistent with belief-state emergence [Hennig et al., 2023].

The role of spatial non-uniformity. The matched-compute control isolates the causal factor. Matched provides more total visual information than foveation but distributes it uniformly, yet its path-history retention ($R^2 = 0.27$ at lag-5) falls below foveated ($R^2 = 0.56$). It is the *spatial non-uniformity* of visual quality, not reduced information volume, that drives compensatory memory.

Why uniform agents have weak memory. Uniform vision yields the weakest spatial encoding across nearly all probes (global GPS $R^2 = 0.614$; path-history collapses to $R^2 = -1.53$ at lag 5). We interpret this as representational outsourcing: when current visual input is reliable, the agent can re-derive spatial information from each frame rather than maintaining it in memory. Foveated and blind agents cannot rely on current visual input and must therefore invest in memory.

A heading-vs-position dissociation in learned gaze. Fov-learned has task performance in the same range as other sighted conditions (SPL 0.820, success 97.8%) but a qualitatively different representation: the strongest compass encoding of any condition ($R^2 = 0.941$, exceeding even the blind agent which receives compass directly) yet chance-level GPS ($R^2 = 0.031$). Only 285/512 units carry > 1 bit of spatial information vs. 461–475 for the other four. This is a categorical shift, not a graded weakening. We believe this traces back to gaze collapse (§4.5): a consistent-view-every-step regime emphasises heading cues (“what wall am I facing”) over position cues (“which room am I in”). Collapsed gaze functions as a head-direction sensor rather than a landmark sensor. Gaze behaviour is thus itself a representational-content axis, which our fixed-gaze conditions cannot access.

5.1 Limitations

(i) Intermediate checkpoints. Foveated-fixed numbers are from a ~ 154 M-frame pre-corruption checkpoint; matched-128 probing is pending convergence. **(ii) Single-seed H3.** Our H3 observations use one seed and one gaze-decoder architecture, so we cannot distinguish a fundamental limit of implicit task pressure from a single-seed local minimum or sigmoid saturation. **(iii) Missing analyses.** Two probes from our plan were not reported: cross-heading generalisation returned insufficient-sample sentinels at our sizes, and a confidence probe (cumulative blur σ) was not implemented; both are in the immediate follow-up. **(iv) Foveation model.** Gaussian blur with quadratic falloff is a simplification; log-polar or multi-scale pyramid models may behave differently. **(v) Linear probes only.** Non-linear probes might reveal structure Ridge misses, particularly for occupancy maps. **(vi) Architecture scope.** The LSTM backbone limits generality to transformer-based architectures where “memory” is defined differently.

6 Conclusion

We trained five navigation agents differing only in visual input and probed their LSTM hidden states with linear decoders, cross-condition CKA, probe transfer, and per-unit spatial information. Three findings emerge. (1) Spatially non-uniform vision induces compensatory temporal memory: the foveated agent retains 56% of its immediate-past encoding quality at 5 steps back, while uniform collapses entirely. (2) No two conditions share a representational geometry (CKA < 0.004 on every pair, catastrophically negative probe transfer) despite all converged sighted conditions surpassing 97% task success. (3) Gaze policy itself acts as a representational-content axis: when gaze collapses, memory encodes heading at the expense of position. Together these establish that *what* a navigation agent’s memory encodes is shaped by the structure of its input — the spatial distribution of visual quality and the policy that directs gaze — not only by the architecture or training objective. Open questions include whether a genuinely epistemic gaze (requiring stochastic or attention-based architectures) recovers or reverses the heading- $\rightarrow$ -position dissociation, and whether the compensatory-memory effect extends to longer episodes and richer sensor topologies.

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A Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation. In the first foveated-learned training run the gaze decoder collapsed (per-dimension $\text{std} \approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \rightarrow 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge. We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma_{\text{target}}^2 - \text{Var}_{\text{batch}}[g])$ where $g \in [0, 1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma_{\text{target}}^2 = 0.01$ ($\text{std} \approx 0.1$), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

Pilot 1 (uncapped, job 2840256). After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0, 1]^2$ ($\text{std} [0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288). Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with $\text{std} [0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation. Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze policy with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

B Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈ 170 M frames of stable progress (reward ≈ 10 , SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL= 0 until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.